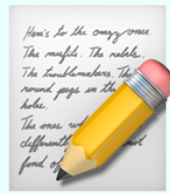


AquaSentinel AI

A Multi-Agent AI Platform for Smart Water Management

Submitted by : Pratham Deshmukh & Aryan Sable



Problem Statement

Challenges Faced by Citizens :

- 💧 Unsafe drinking water
- 🚰 Choosing the right purifier
- 🌱 Water wastage
- 📋 Complicated complaint registration

Why it matters

- 📌 Information is scattered across multiple platforms.
- 🚰 Complaint reporting is slow and inconvenient.
- 🤖 Lack of a single intelligent assistant.

🔵 **Problem:** Water-related information and services are fragmented, making it difficult for citizens to get timely guidance and report issues efficiently.

🌍 **WHO:** Billions of people still lack access to safely managed drinking water.

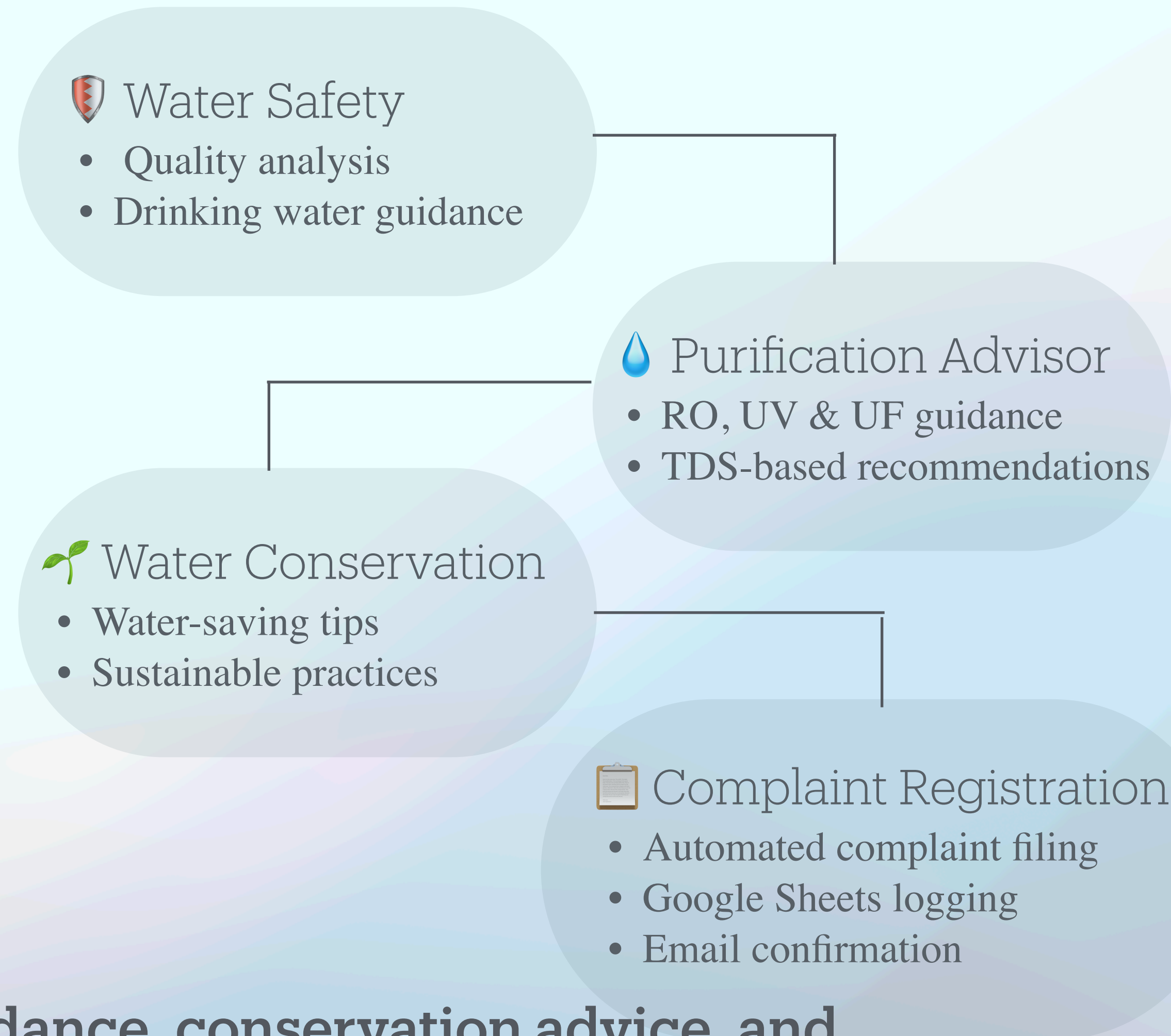


Introducing AquaSentinel AI

An Intelligent Multi-Agent AI System for Smart Water Assistance

What is AquaSentinel AI ?

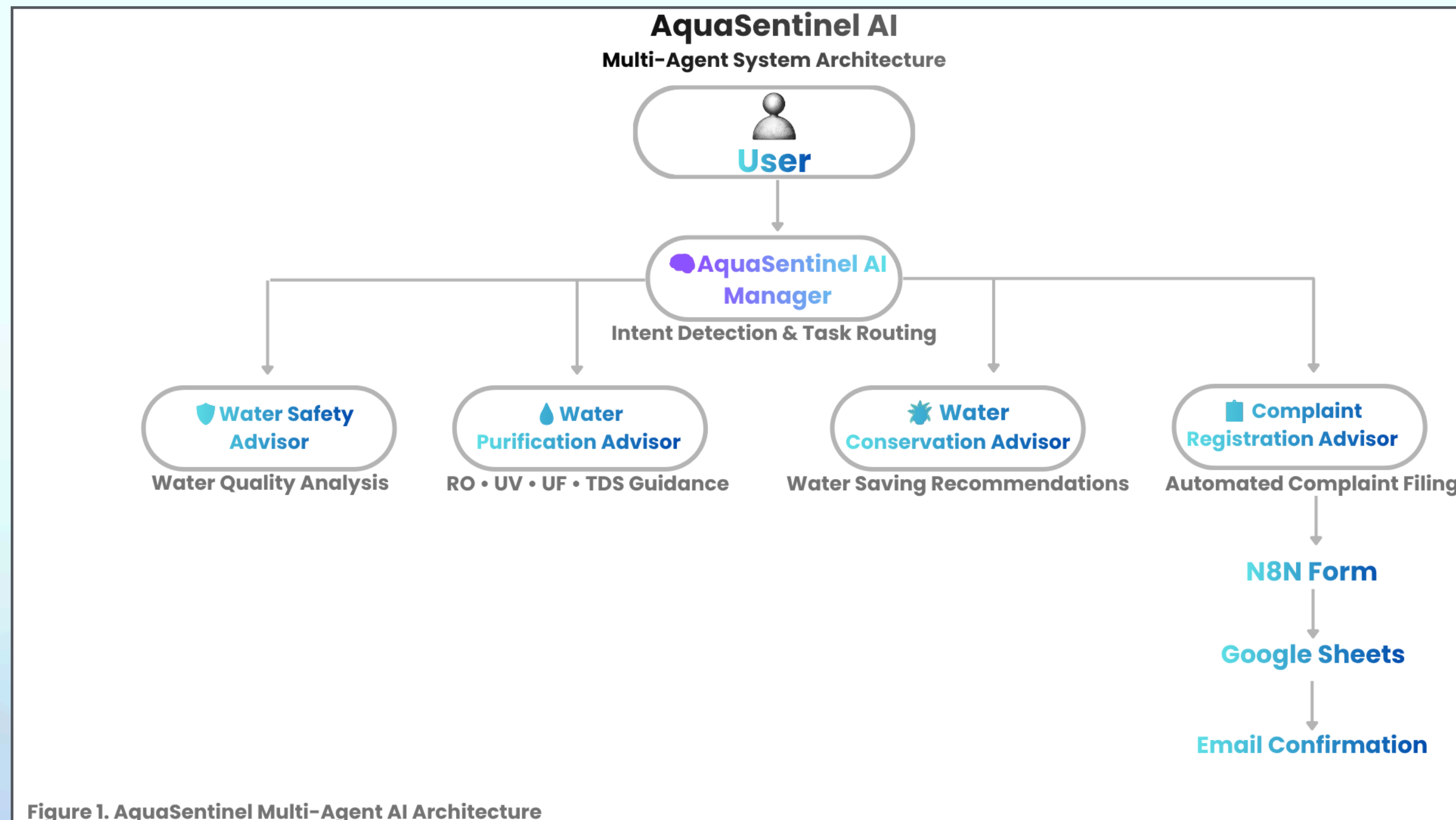
AquaSentinel AI is a multi-agent AI platform that helps citizens by providing reliable water-related guidance through specialized AI advisors. It understands the user's request, routes it to the appropriate expert, and automates complaint registration when required.



 AquaSentinel AI brings water safety, purification guidance, conservation advice, and complaint registration together in one intelligent platform.

System Architecture

How AquaSentinel AI Processes User Requests



User Interaction

- User enters a water-related query.
- The AI Manager identifies the user's intent.
- The request is routed to the appropriate specialist.

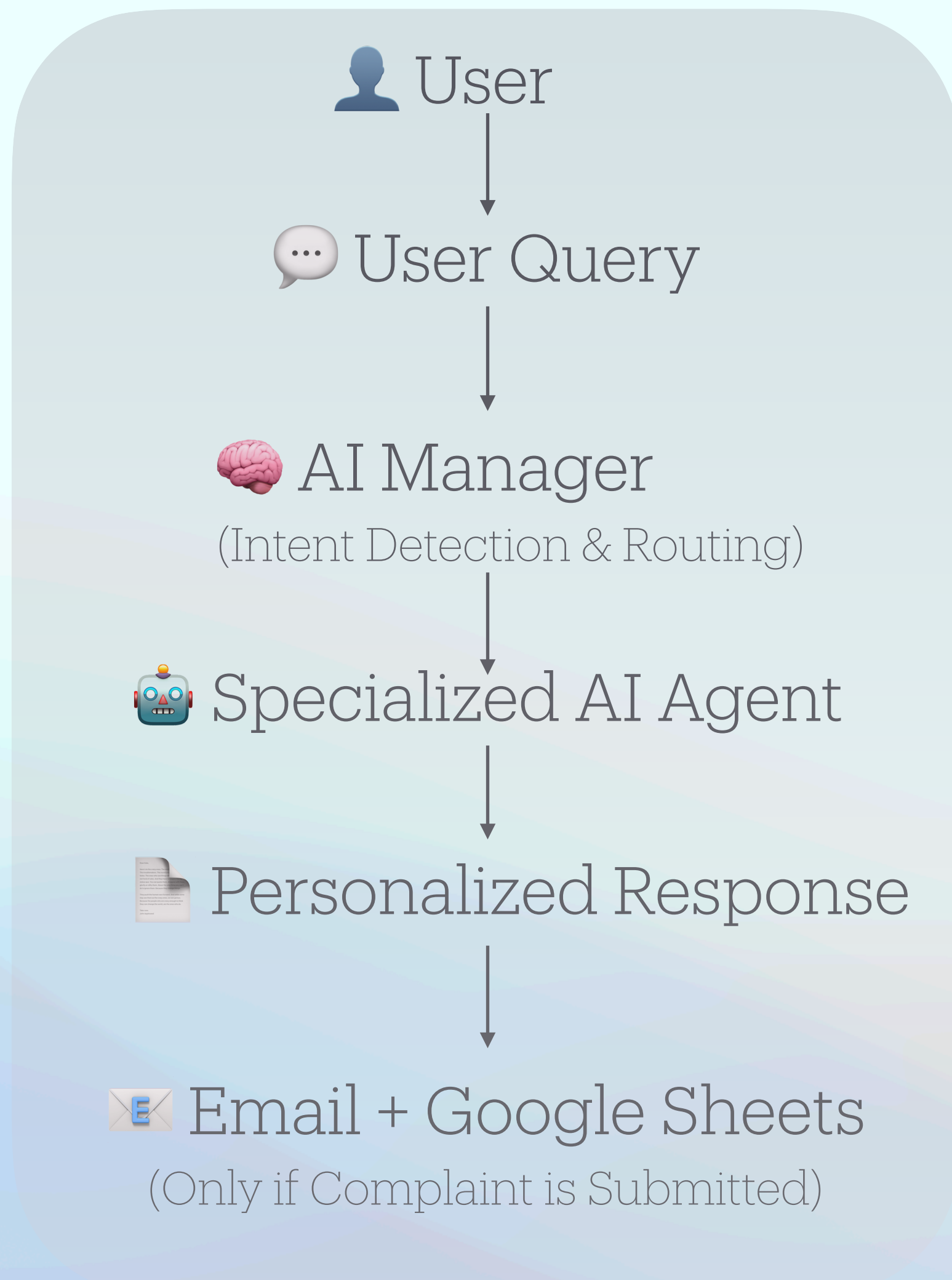
Response Generation

- Specialist AI provides domain-specific guidance.
- Complaint requests are sent to the Complaint Registration module.
- Complaint data is stored in Google Sheets and a confirmation email is sent.

AI Manager intelligently routes every query to the right specialist, making responses more accurate and efficient.

Project Workflow

How AquaSentinel AI Handles a User Query



Real-world Example →

User Query: "My tap water is brown. I want to report it."

AI Manager identifies:

- Water Safety
- Complaint Registration

Water Safety Advisor provides safety recommendations.

Complaint Registration Advisor opens the complaint form.

Complaint is stored in Google Sheets.

Confirmation email is sent.

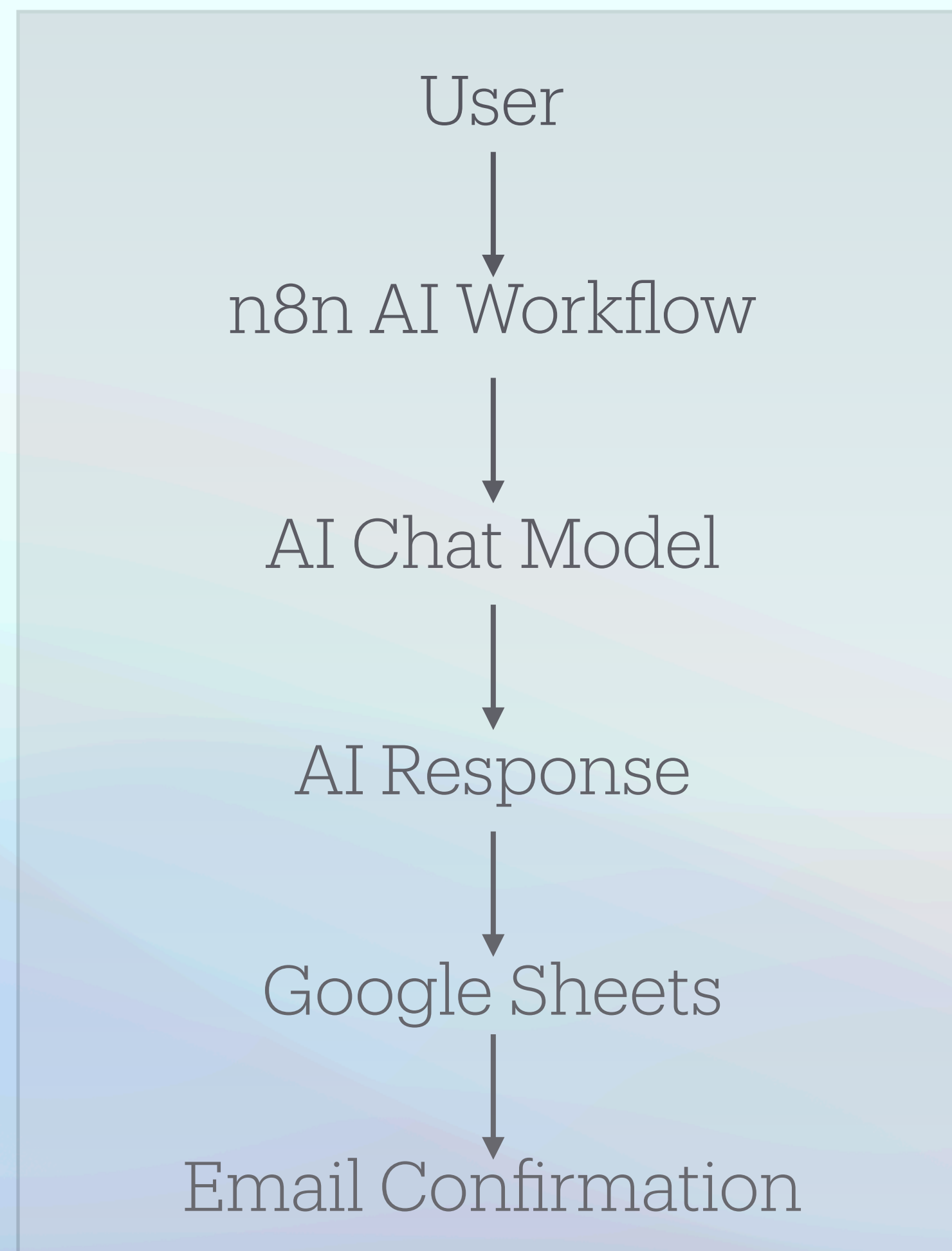
 By combining multiple AI specialists into one workflow, AquaSentinel AI makes water-related assistance simple, accessible, and efficient for everyone.



Technology Stack

Technologies used to build AquaSentinel AI

Workflow Pipeline



Core Technologies



n8n

- Workflow Automation
- AI Agent Orchestration



Google Sheets

- Complaint Database
- Data Logging



n8n Chat

- User Interaction
- Complaint Form



Groq /Gemini API

- Llama 3 Model
- Fast AI Responses



Gmail

- Automatic Email Confirmation



GitHub

- Version Control
- Project Repository



Live Example

How AquaSentinel AI Solves a Real User Problem

User asks

"My tap water is brown and smells like chlorine. My TDS is around 650. Is it safe to drink?
Should I buy a purifier? I also want to report this issue."



AquaSentinel AI Manager

Detects multiple intents and routes
the request to the correct AI specialists.

(Multi-Agent Decision)



Water Safety Advisor

- ✓ Explains possible causes
- ✓ Suggests immediate precautions



Purification Advisor

- ✓ Recommends RO purifier
- ✓ Explains why based on TDS



Complaint Advisor

- ✓ Opens complaint form
- ✓ Stores details in Google Sheets
- ✓ Sends confirmation email



Result

The user receives safety advice, purifier recommendations, and completes complaint registration in a single conversation.

✨ Key Features & Benefits

🚀 What Makes AquaSentinel AI Useful?

🛡️ Water Safety Analysis

Provides quick guidance on drinking water quality.

- Identifies possible water quality concerns
- Suggests immediate safety precautions
- Helps users make informed decisions

💧 Personalized Purification Guidance

Recommends the most suitable purifier based on water conditions.

- Uses TDS values for recommendations
- Explains RO, UV, and UF technologies
- Helps users avoid unnecessary expenses

📋 Smart Complaint Registration

Simplifies reporting water-related issues.

- Generates complaint forms automatically
- Stores records in Google Sheets
- Sends instant email confirmation

🧠 Intelligent Multi-Agent System

Routes every request to the right AI specialist.

- Detects user intent automatically
- Assigns tasks to dedicated AI advisors
- Delivers focused and accurate responses

🌍 Real-World Impact

- ✓ Makes water-related information easier to access
- ✓ Encourages safe water practices and public awareness
- ✓ Designed for future expansion with regional languages and voice interaction
- ✓ Reduces confusion by combining multiple services into one platform



Future Scope

Future enhancements to make water assistance smarter and more accessible.

Regional Language Support

- Support Marathi, Hindi, Tamil and other Indian languages
- Make the platform accessible to more citizens



Voice-Based Interaction

- Allow users to speak instead of typing
- Helpful for elderly and less literate users



GPS-Based Complaint Registration

- Automatically capture user location
- Speed up issue reporting to local authorities



Smart Water Analytics

- Analyze complaint trends
- Identify high-risk areas
- Help authorities make better decisions

Long-Term Vision

Our vision is to develop AquaSentinel AI into a smart public assistant that helps citizens access safe water information, report issues efficiently, and supports authorities with better decision-making.

★ Conclusion & Thank You

AquaSentinel AI – Smarter Water Assistance Through Multi-Agent AI

🎯 Project Summary

AquaSentinel AI combines specialized AI agents to provide reliable water guidance, purifier recommendations, water conservation tips, and automated complaint registration through a single intelligent platform.

🌍 Why It Matters

- 💧 Promotes safe water awareness
- 🤖 Makes water-related assistance simple and accessible
- 📋 Simplifies complaint reporting through automation
- 🌱 Designed for future expansion with regional languages and voice interaction

📁 Project Resources

GitHub Repository

<https://github.com/prathamdesh07-afk/AquaSentinel-AI>



📱 Scan to view source code
& documentation

Thank you for your time! We welcome your questions and feedback.